

Best Next Architecture for Interpretable Drug-Combination Synergy Prediction

Bottom line

Given your exact regime—two drugs plus one cell line, a DeepSynergy-style baseline, only 59 unique cell lines, harder OOD-style splits, and a serious need for reverse engineering at the gene level—the best next step is **not** a from-scratch gene-token transformer ¹, a cell-line graph neural network ², or a generic autoencoder trained only on the synergy data. The strongest next architecture for the **cell-line branch** is a **knowledge-guided sparse encoder** that maps expression into a compact set of biologically named program scores—pathway activities and/or transcription-factor activities—and then passes those scores through a **small supervised MLP bottleneck** to produce the final cell embedding. In practice, that means replacing your current `genes -> PCA -> latent` pipeline with `genes -> pathway/regulon activity layer -> small bottleneck`, and treating the structured intermediate layer as the main object of interpretation. That recommendation fits the strongest current evidence: prior biological feature selection helps synergy models, fully connected feature-encoding subnetworks remain very competitive or best in systematic evaluations, and modern synergy models often exploit repeated drug or cell identities rather than learning genuinely transferable biology under strict splits. ³

The second important conclusion is that your current challenge is mostly about **inductive bias**, not raw model capacity. The literature on DrugComb ⁴-style and NCI-ALMANAC ⁵-style evaluation now makes it hard to defend “just use a bigger encoder” as a next step: recent benchmark work found poor generalization to unseen drugs and unseen cell lines, and found that biologically meaningful drug or cell-line features often did not drive predictive performance as much as people expected. That makes your harder split design exactly the right lens through which to choose the next architecture. ⁶

If, later, you bring in substantial external transcriptomic or perturbation data, the ranking changes. At that point, pre-trained deconfounding or contrastive encoders, ontology-guided latent models, and eventually foundation-model-style cell encoders become much more attractive. But that is a **later** phase. For the next clean experiment, the right move is a structured, interpretable, low-capacity cell encoder rather than a high-capacity universal one. ⁷

What your regime is really telling you

Your current principal component analysis ⁸ pipeline is already revealing the central fact of the project: the raw 23.8k-dimensional expression vector is too unconstrained for the amount of real cellular diversity you have, so the performance gain from PCA is mostly telling you that **aggressive regularization and feature compression are necessary**. With only 59 unique cell lines, the problem is not “how do I learn a richer cell manifold?” so much as “what coordinate system lets the model use a few dozen biological contexts efficiently without destroying interpretability?” That is why PCA helped even though it made biological explanation harder. ⁹

The recent synergy evaluation literature makes the stakes of that choice very clear. SynVerse reports that biologically meaningful drug or cell-line features and drug-drug interaction features often did **not**

drive predictive performance, and that all tested models generalized poorly to unseen drugs and cell lines. The 2026 “One-hot News” study goes further: replacing chemical and biological representations with simple one-hot identifiers produced comparable or slightly better performance across several published models, and the authors argue that repeating drug and cell-line structure impairs true feature-based learning. A recent review reaches the same broad conclusion: current methods do much better when some drugs or cell lines are already known than in scenarios involving genuinely new drugs or new cellular contexts. In other words, your harder splits are not just stricter; they are closer to the real scientific question. ¹⁰

The same body of evidence also says something useful about your interpretability goal. In a systematic evaluation on ALMANAC, feature selection based on prior biological knowledge improved performance, and fully connected feature-encoding subnetworks outperformed the more elaborate alternatives that were tested. That is important for your decision: if a simpler structured bottleneck can regularize the cell branch *and* expose interpretable coordinates, you should prefer that over a more flexible architecture whose apparent sophistication is not supported by this sample regime. ¹¹

Candidate architectures compared

Knowledge-guided pathway or regulon encoder plus small MLP. This is the option I would rank first. By this I mean a fixed or lightly trainable sparse gene-to-program layer built from resources such as PROGENy for pathway activity and DoRothEA for transcription-factor activity, followed by a small bottleneck MLP. PROGENy was designed specifically to summarize high-dimensional transcriptomics into a small number of pathway activities, and DoRothEA benchmarked transcription-factor regulons for activity inference. In synergy and drug-response work, biologically guided feature selection has repeatedly helped compared with using the full raw transcriptome. For your project, the upside is unusually strong: OCA becomes directly meaningful at the pathway or TF level; gene-level back-mapping is straightforward because every latent coordinate is attached to a defined gene set; the representation is compact without inventing arbitrary PCA axes; and the branch is sample-efficient because the biological structure is mostly fixed rather than learned from 59 cell lines. The main downside is rigidity: if the pathway vocabulary is too coarse or too incomplete, you can lose predictive signal. That caution is real, because benchmark studies of interpretable drug-response models found that **explicit** pathway-latent models often underperformed strong black-box MLPs, and some apparent gains could be reproduced with random pathway assignments. The implication is not “do not use pathway structure.” It is “use pathway structure as a compact feature basis, not as a very deep hand-crafted visible network unless you validate it aggressively.” ¹²

Generic autoencoder or variational autoencoder ¹³. From scratch on your current synergy data, I would rank this clearly below the pathway/regulon encoder. A generic AE or VAE will optimize reconstruction, not synergy. Recent drug-response work explicitly notes that AE-derived cell embeddings may preserve transcriptomic structure while missing the information that matters most for response prediction, and contrastive alternatives were proposed partly for that reason. The picture changes if you add external data or biological constraints. DIPK improved generalization to novel cells and drugs by integrating gene interaction priors and self-supervised components; CODE-AE improved OOD robustness by separating intrinsic biological signal from context-specific confounding; and ontology-guided models such as OntoVAE can align latent variables with pathway or phenotype activities and support perturbation-style in silico analyses. So the nuanced answer is: **a biologically constrained or externally pretrained AE/VAE is promising; a generic AE/VAE trained only on 59 cell lines is not the right next move.** ¹⁴

Cell-line gene graph model. A cell-line GNN becomes interesting only when you are willing to make the graph itself a first-class modeling object. There is respectable evidence in drug response and synergy that graph-based priors can help: TranSynergy uses propagated drug-target and gene-interaction information, DIPK builds on integrated gene networks, and newer contextual graph models try to make the cellular network itself condition-specific. The benefit is conceptual coherence: message passing can encode gene-gene dependency structure and may produce more mechanism-like representations than raw vectors. But the cost in your setting is high. With bulk expression and only 59 unique cell lines, the graph prior risks becoming more important than the actual data. Interpretation also becomes conditional on graph construction choices, coverage, confidence thresholds, and propagation rules. Since benchmark work already warns that explicit biologically structured latent layers do not automatically outperform tuned black-box baselines, I would treat a cell GNN as a **later-stage research direction**, not the next pragmatic upgrade. It becomes more defensible if you later add strong priors such as drug targets, CRISPR dependencies, or cell-specific interactomes. ¹⁵

From-scratch transformer over genes. This is the most obvious “same-family-for-both-modalities” answer, because the same broad idea can encode SMILES and gene tokens. It is also the wrong next answer for your current project. The strong transformer results in biology come from pretraining at much larger scale: Geneformer was pretrained on roughly 30 million single cells, scGPT on millions of single-cell profiles, and XPert-style perturbation models leverage large perturbation corpora. Outside biology, deep tabular benchmarks also do not show transformers to be universally superior to simpler tuned architectures. Put differently: transformer-style models can be excellent **after** pretraining, but your present dataset is not big enough for the cell branch to teach one a useful gene-language from scratch. Relative to the structured bottleneck alternative, a transformer now would cost more, explain less directly, and still be underdetermined by the number of unique cell lines you actually have. ¹⁶

Hybrid encoder. If you are worried that a pure pathway/regulon encoder will leave predictive signal on the table, the strongest compromise is a **hybrid**: one interpretable program branch, plus one very small residual branch over a biologically selected gene subset. The interpretable branch carries the biology you plan to reverse engineer; the residual branch is just a safety valve for signal the curated programs miss. This compromise is consistent with recent evidence that generic AE latents are often not task-optimal, while task-aware contrastive or self-supervised encoders can produce more useful cell representations and a more balanced reliance on drug and cell features. The danger is obvious: if the free branch becomes large, the model will retreat into an uninterpretable solution. So I would only use this hybrid if the pathway-only model loses a non-trivial amount of accuracy. ¹⁷

If I turn that comparison into a practical ranking **for your current regime**, it is: **pathway/regulon encoder first; externally pretrained or ontology-guided AE/VAE second; cell GNN third; from-scratch transformer last**. The hybrid option sits between the first and second positions as a practical compromise if the fully interpretable version gives up too much predictive power. ¹⁸

Branch design across drugs and cell lines

You should **keep separate architectures for the drug and cell branches**. The right symmetry in a two-drug model is not “drug and cell should share the same family”; it is “drug A and drug B should be encoded symmetrically, usually with shared weights.” That is already how successful multimodal models tend to think about the problem. MatchMaker uses two drug-specific subnetworks that each combine one drug with the same cell context before fusion; DeepDDS uses a graph-based drug encoder but not a graph-based cell encoder; TranSynergy combines drug-target/network information with cell-line gene features in a modality-specific way. The common design pattern is heterogeneous encoders with a shared prediction head, not one universal encoder family across all modalities. ¹⁹

There is also a strategic experimental reason to keep the branches separate right now: you want to understand whether the **cell representation** is helping. If you change both the drug branch and the cell branch at once, you lose attribution of the gain. The clean next experiment is therefore: keep your current hashed SMILES branch and current fusion head, replace only the cell encoder, and measure what changes under random, drug, cell-line, and combined-like stricter splits. After that, and especially for the hardest OOD regime, the drug branch absolutely deserves attention too. A systematic ALMANAC evaluation found drug features more predictive than omics features, and simple fingerprint-based drug representations slightly outperformed learned drug representations in that benchmark, even though models like DeepDDS show that molecular graphs can also work well. So the long-run order should be: **first isolate the cell-branch upgrade; then compare the current drug representation against strong fixed chemical baselines and a molecular GNN or pretrained molecular encoder.** ²⁰

If you absolutely forced one common family across modalities, the only defensible answer would be a transformer-style setup, because tokenized SMILES and gene tokens can both be fed to attention-based encoders. I still would not do that now. In your current regime, architectural consistency is less important than **consistent interfaces**: each branch should output a normalized embedding of similar scale and similar downstream dimensionality, but they do not need to share the same upstream machinery. ²¹

If the task shifts to one drug and one cell line

If you later pivot from two drugs plus one cell line to **one drug plus one cell line**, the answer changes less than it may seem. The **cell-line encoder recommendation stays basically the same**: I would still use a pathway or regulon activity encoder plus a small bottleneck. In fact, the interpretability story gets cleaner, because you no longer need to disentangle baseline cell biology from the interaction of two separate drug branches. You can ask a more direct question: which cell-state programs make this one drug more or less effective? That makes drug-conditioned gating or low-rank bilinear fusion over pathway activities especially attractive in the monotherapy case. ²²

What does change is the value of pretraining and transfer. In the one-drug setting, monotherapy response datasets become directly usable, and recent work is already transferring knowledge from monotherapy into synergy prediction with interpretable multitask designs. That means the externally pretrained hybrid route moves **up** in the ranking if your future program includes monotherapy: pretrained cell encoders, multitask drug-response heads, and contrastive drug or cell representations become materially more attractive. So the practical answer is: **the immediate cell encoder does not really change, but the medium-term roadmap becomes more transfer-learning-heavy once you move to one-drug response.** ²³

Recommended build and roadmap

If I had to recommend **one practical next architecture right now**, it would be this: **z-scored expression -> fixed sparse gene-to-program layer -> small supervised MLP bottleneck -> normalized cell embedding**, with the gene-to-program layer built from compact pathway and regulon activity definitions rather than thousands of raw genes. A very practical instantiation is to compute PROGENy pathway activities and DoRoThEA TF activities, concatenate them, apply LayerNorm, pass them through a 2-layer MLP, and output a 64- to 128-dimensional cell embedding. If your downstream fusion code strongly expects 128–256 dimensions, keep the **interpretable** layer in the tens-to-low-hundreds of biological programs, then project to 128 with one final linear layer; do **not** confuse that projection with the biologically meaningful representation. Keep the drug branches and fusion head unchanged in this experiment, add per-branch normalization so the cell vector cannot dominate by

scale alone, and run OCA first on pathways or TFs and only then on genes within the top-ranked programs. Also report seed-wise and fold-wise attribution stability, plus shuffled-program controls, because the reliability literature on biology-inspired networks shows that interpretation stability is not automatic. ²⁴

Best practical next step: pathway or regulon activity encoder plus small bottleneck MLP, with a very small residual selected-gene branch only if the interpretable version loses too much accuracy. As a sanity baseline, I would also run one simpler model with no pathway structure at all: a biologically selected gene subset—such as landmark or drug-response-related genes—fed directly into a small residual MLP. The literature strongly supports biological feature selection, and many synergy models already work with curated gene subsets instead of the whole transcriptome. ²⁵

Best interpretable next step: a visible sparse gene-to-program network, but keep it shallow and validate it aggressively. If you want a larger pathway catalog rather than footprint or regulon scores, Reactome is the strongest default among standard pathway collections in DRP benchmarking, but you should still compare it against random-pathway controls and repeated training seeds before trusting any mechanistic story. ²⁶

Best long-term research direction: a modular multimodal model with (i) a stronger drug branch—at least against Morgan or ECFP fingerprints, and then possibly a molecular GNN or pretrained molecular language model—plus (ii) a cell branch pretrained on large external transcriptomic or perturbation data using deconfounding, contrastive, ontology-guided, or foundation-model-style representation learning, and (iii) a multitask objective linking monotherapy and synergy. That is the route most likely to help the hardest OOD splits. It is also the route most likely to make sense if your program later expands from synergy prediction into monotherapy or perturbation-response modeling. ²⁷

In short: **for this project, right now, choose a biologically structured cell encoder, not a bigger universal one.** The model should learn in a coordinate system that a biologist can name, that OCA can probe directly, and that a dataset with only 59 unique cell lines can actually support.

¹ ¹⁴ ¹⁷ <https://www.nature.com/articles/s41698-024-00589-8>
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